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Inventor(s)	Sayed; Ashley John et al.

Aerosol provision device

Abstract

An aerosol provision device comprises a haptic component configured to provide haptic feedback, a battery configured to power the haptic component, and an electrically insulating resilient member in contact with the haptic component and positioned between the battery and the haptic component.

Inventors:	Sayed; Ashley John (London, GB), Thorsen; Mitchel (Madison, WI), Warren; Luke James (London, GB)
Applicant:	Nicoventures Trading Limited (London, GB)
Family ID:	70050026
Assignee:	Nicoventures Trading Limited (London, GB)
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Primary Examiner: Chambers; Travis S

Attorney, Agent or Firm: MERCHANT & GOULD P.C.

Background/Summary

PRIORITY CLAIM (1) The present application is a National Phase entry of PCT Application No. PCT/EP2020/056248, filed Mar. 9, 2020, which claims priority from U.S. Provisional Application No. 62/816,326, filed Mar. 11, 2019, each of which is hereby fully incorporated by reference herein.

TECHNICAL FIELD

(1) The present invention relates to an aerosol provision device.

BACKGROUND

(2) Smoking articles such as cigarettes, cigars and the like burn tobacco during use to create tobacco smoke. Attempts have been made to provide alternatives to these articles that burn tobacco by creating products that release compounds without burning. Examples of such products are heating devices which release compounds by heating, but not burning, the material. The material may be for example tobacco or other non-tobacco products, which may or may not contain nicotine.

SUMMARY

(3) According to an aspect of the present disclosure, there is provided an aerosol provision device, comprising: a haptic component configured to provide haptic feedback; a battery configured to power the haptic component; and an electrically insulating, resilient member positioned between the battery and the haptic component.

(4) Further features and advantages of the invention will become apparent from the following description of preferred embodiments of the invention, given by way of example only, which is made with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention will now be described, by way of example only, with reference to the accompanying drawings, in which:

(2) FIG. 1 shows a front view of an example of an aerosol provision device;

(3) FIG. 2 shows a front view of the aerosol provision device of FIG. 1 with an outer cover removed;

(4) FIG. 3 shows a cross-sectional view of the aerosol provision device of FIG. 1;

(5) FIG. 4 shows an exploded view of the aerosol provision device of FIG. 2;

(6) FIG. 5A shows a cross-sectional view of a heating assembly within an aerosol provision device;

(7) FIG. 5B shows a close-up view of a portion of the heating assembly of FIG. 5A;

(8) FIG. 6A shows a perspective view of a battery support and battery;

(9) FIG. 6B shows a close-up of a portion of FIG. 6A;

(10) FIG. 7 shows a side view of an example resilient member;

(11) FIG. 8 shows a perspective view of the example resilient member of FIG. 7;

(12) FIG. 9 shows a perspective view of the example resilient member of FIG. 7 engaged with a haptic component; and

(13) FIG. 10 shows a perspective view of the example resilient member of FIG. 7 engaged with a battery.

DETAILED DESCRIPTION OF THE DRAWINGS

(14) As used herein, the term “aerosol generating material” includes materials that provide volatilized components upon heating, typically in the form of an aerosol. Aerosol generating material includes any tobacco-containing material and may, for example, include one or more of tobacco, tobacco derivatives, expanded tobacco, reconstituted tobacco or tobacco substitutes. Aerosol generating material also may include other, non-tobacco, products, which, depending on the product, may or may not contain nicotine. Aerosol generating material may for example be in the form of a solid, a liquid, a gel, a wax or the like. Aerosol generating material may for example also be a combination or a blend of materials. Aerosol generating material may also be known as “smokable material”.

(15) Apparatuses are known that heat aerosol generating material to volatilize at least one component of the aerosol generating material, typically to form an aerosol which can be inhaled, without burning or combusting the aerosol generating material. Such apparatus is sometimes described as an “aerosol generating device,” an “aerosol provision device,” a “heat-not-burn device,” a “tobacco heating product device,” or a “tobacco heating device” or similar. Similarly, there are also so-called e-cigarette devices, which typically vaporize an aerosol generating material in the form of a liquid, which may or may not contain nicotine. The aerosol generating material may be in the form of or be provided as part of a rod, cartridge or cassette or the like which can be inserted into the apparatus. A heater for heating and volatilizing the aerosol generating material may be provided as a “permanent” part of the apparatus.

(16) An aerosol provision device can receive an article comprising aerosol generating material for heating. An “article” in this context is a component that includes or contains in use the aerosol generating material, which is heated to volatilize the aerosol generating material, and optionally other components in use. A user may insert the article into the aerosol provision device before it is heated to produce an aerosol, which the user subsequently inhales. The article may be, for example, of a predetermined or specific size that is configured to be placed within a heating chamber of the device which is sized to receive the article.

(17) A first aspect of the present disclosure defines an aerosol provision device comprising a haptic component configured to provide haptic feedback, such as a vibration, and a battery configured to power the haptic component. Positioned between the battery and the haptic component is an electrically insulating resilient member. The resilient member electrically insulates the haptic component from the battery to avoid shorting the battery. The resilient member therefore acts as a barrier to stop the battery (or another electrically conductive member connected to the battery) from contacting the haptic component.

(18) In some examples, the resilient member is in contact with the haptic component. In certain examples, the resilient member is also in contact with the battery. By being in contact with the haptic component the resilient member helps transfer vibrations from the haptic component to other components of the device, such as an outer cover which is held by a user. This helps ensure that a user of device is able to feel the vibrations. For example, the resilient member may reduce vibrations which are transmitted to the battery and increase vibrations in directions away from the battery. By being resilient and in contact with the haptic component, the resilient member can apply a force to the haptic component to transfer vibrations away from the battery. A resilient member can be deformed.

(19) The aerosol provision device may further comprise a heater assembly, which may include at least one inductor coil. The heater assembly may also comprise a susceptor. The battery may also power the heater assembly.

(20) In certain examples the resilient member is a dielectric. The resilient member may have a resistivity of between about 4×10^{15} Ohm-cm and about 6×10^{15} Ohm-cm, such as about 5×10^{15} Ohm-cm.

(21) The resilient member may comprise silicone (i.e. the resilient member may be made from a polymerized siloxane). Silicone is a good electrical insulator, is resilient, and is a good insulator of heat. In a particular example, the silicone is silicone rubber.

(22) The device may further comprise an electrically conductive member in contact with a battery terminal of the battery, and the resilient member may be in contact with the electrically conductive member. The electrically conductive members may be wires or conductive strips which connect the battery to the PCB and/or other components of the device. The terminal may be a positive or negative terminal. Thus, the resilient member electrically insulates the haptic component from the electrically conductive member. In some arrangements the resilient member is in contact with the electrically conductive member and the battery.

(23) The resilient member may be positioned between a battery terminal and the haptic component. The resilient member therefore additionally or alternatively insulates the haptic component from the battery terminal. The battery terminal may be arranged at one end of the battery for example.

(24) The resilient member and the haptic component may each comprise corresponding engagement features for maintaining a relative position between the resilient member and the haptic component. The engagement features therefore help keep the resilient member and haptic component in contact with each other because vibrations can otherwise cause the resilient member and haptic component to move relative to each other.

(25) In a specific arrangement, the resilient member may comprise a protrusion as a type of engagement feature, and the haptic component may comprise a recess as a corresponding engagement feature, which receives the protrusion. Alternatively, the haptic component may comprise a protrusion, and the resilient member comprises a recess, which receives the protrusion. Other types of engagement features may be used instead.

(26) The resilient member may be at least partially compressed, thereby to exert a force on the haptic component away from the battery. For example, the battery and haptic component may be arranged such that the resilient member is squeezed between them, which causes the haptic component to be biased, or subject to a force, away from the resilient member. This causes the haptic component to be forced against other components of the device, which helps transfer vibrations.

(27) For example, the device may comprise an end member which defines at least a portion of an outer surface of the device, wherein the haptic component is arranged between the end member and the resilient member, and the resilient member forces the haptic component towards the end member. One or more other components may be arranged between the haptic component and the end member.

(28) The resilient member may comprise a first portion and a second portion, the first portion being thicker than the second portion, wherein the haptic component is in contact with the first portion, and the second portion is in contact with at least one of: (i) an electrically conductive member in contact with a battery terminal, and (ii) a battery terminal. The thicker first portion acts to provide a biasing force to the haptic component so that the vibrations are transferred away from the battery and towards other components of the device. The thinner second portion can be thinner because it provides electrical insulation. The thickness of the first and second portions are measured in a direction parallel to a longitudinal axis of the device, which is also parallel to a direction away from the battery towards the haptic component.

(29) The second portion may comprise a recess configured to receive an end of the battery. The recess may also be known as a receptacle. The recess may be defined by one or more side walls and a base, where the base receives the end of the battery and the one or more side walls abut a side surface of the battery. Preferably the recess is shaped to conform to the shape of the battery. The

recess helps hold the resilient member in position relative to the battery. Instead of a recess, the second portion may comprise an engagement feature, and the battery may comprise a corresponding engagement feature, where the engagement features are for maintaining a relative position between the resilient member and the battery.

(30) The resilient member may be at least 0.5 mm thick between the battery and the haptic component. In a particular arrangement the resilient member is less than about 6 mm thick. The thickness of the resilient member is the smallest measurable distance between the battery and the haptic component. It has been found that these dimensions provide a good balance between providing adequate electrical insulation and a strong biasing force, while also ensuring suitable device dimensions.

(31) In one example, the first portion has a thickness of between about 2 mm and about 3 mm, the second portion has a thickness of between about 0.5 mm and about 2 mm. Preferably the first portion has a thickness of between about 2 mm and about 2.5 mm. The thickness of the second portion is the minimum distance between the first portion and the battery.

(32) If the second portion defines a recess, preferably the first portion has a thickness of between about 2 mm and about 3 mm, the base of the second portion has a thickness of between about 0.5 mm and about 2 mm, and the side walls have a thickness of about 1 to about 3 mm.

(33) If the second portion does not define a recess, the first portion may have a thickness of between about 2 mm and about 3 mm, and the second portion may have a thickness of between about 0.5 mm and about 2 mm, more preferably between about 0.5 mm and about 1 mm.

(34) Preferably, the device is a tobacco heating device, also known as a heat-not-burn device.

(35) FIG. 1 shows an example of an aerosol provision device **100** for generating aerosol from an aerosol generating medium/material. In broad outline, the device **100** may be used to heat a replaceable article **110** comprising the aerosol generating medium, to generate an aerosol or other inhalable medium which is inhaled by a user of the device **100**.

(36) The device **100** comprises a housing **102** (in the form of an outer cover) which surrounds and houses various components of the device **100**. The device **100** has an opening **104** in one end, through which the article **110** may be inserted for heating by a heating assembly. In use, the article **110** may be fully or partially inserted into the heating assembly where it may be heated by one or more components of the heater assembly.

(37) The device **100** of this example comprises a first end member **106** which comprises a lid **108** which is moveable relative to the first end member **106** to close the opening **104** when no article **110** is in place. In FIG. 1, the lid **108** is shown in an open configuration, however the lid **108** may move into a closed configuration. For example, a user may cause the lid **108** to slide in the direction of arrow "A".

(38) The device **100** may also include a user-operable control element **112**, such as a button or switch, which operates the device **100** when pressed. For example, a user may turn on the device **100** by operating the switch **112**.

(39) The device **100** may also comprise an electrical component, such as a socket/port **114**, which can receive a cable to charge a battery of the device **100**. For example, the socket **114** may be a charging port, such as a USB charging port.

(40) FIG. 2 depicts the device **100** of FIG. 1 with the outer cover **102** removed and without an article **110** present. The device **100** defines a longitudinal axis **134**.

(41) As shown in FIG. 2, the first end member **106** is arranged at one end of the device **100** and a second end member **116** is arranged at an opposite end of the device **100**. The first and second end members **106**, **116** together at least partially define end surfaces of the device **100**.

(42) For example, the bottom surface of the second end member **116** at least partially defines a bottom surface of the device **100**. Edges of the outer cover **102** may also define a portion of the end surfaces. In this example, the lid **108** also defines a portion of a top surface of the device **100**.

(43) The end of the device closest to the opening **104** may be known as the proximal end (or mouth

end) of the device **100** because, in use, it is closest to the mouth of the user. In use, a user inserts an article **110** into the opening **104**, operates the user control **112** to begin heating the aerosol generating material and draws on the aerosol generated in the device. This causes the aerosol to flow through the device **100** along a flow path towards the proximal end of the device **100**.

(44) The other end of the device furthest away from the opening **104** may be known as the distal end of the device **100** because, in use, it is the end furthest away from the mouth of the user. As a user draws on the aerosol generated in the device, the aerosol flows away from the distal end of the device **100**.

(45) The device **100** further comprises a power source **118**. The power source **118** may be, for example, a battery, such as a rechargeable battery or a non-rechargeable battery. Examples of suitable batteries include, for example, a lithium battery (such as a lithium-ion battery), a nickel battery (such as a nickel-cadmium battery), and an alkaline battery. The battery is electrically coupled to the heating assembly to supply electrical power when required and under control of a controller (not shown) to heat the aerosol generating material. In this example, the battery is connected to a central support **120** which holds the battery **118** in place. The central support **120** may also be known as a battery support, or battery carrier.

(46) The device further comprises at least one electronics module **122**. The electronics module **122** may comprise, for example, a printed circuit board (PCB). The PCB **122** may support at least one controller, such as a processor, and memory. The PCB **122** may also comprise one or more electrical tracks to electrically connect together various electronic components of the device **100**. For example, the battery terminals may be electrically connected to the PCB **122** so that power can be distributed throughout the device **100**. The socket **114** may also be electrically coupled to the battery via the electrical tracks.

(47) In the example device **100**, the heating assembly is an inductive heating assembly and comprises various components to heat the aerosol generating material of the article **110** via an inductive heating process. Induction heating is a process of heating an electrically conducting object (such as a susceptor) by electromagnetic induction. An induction heating assembly may comprise an inductive element, for example, one or more inductor coils, and a device for passing a varying electric current, such as an alternating electric current, through the inductive element. The varying electric current in the inductive element produces a varying magnetic field. The varying magnetic field penetrates a susceptor suitably positioned with respect to the inductive element, and generates eddy currents inside the susceptor. The susceptor has electrical resistance to the eddy currents, and hence the flow of the eddy currents against this resistance causes the susceptor to be heated by Joule heating. In cases where the susceptor comprises ferromagnetic material such as iron, nickel or cobalt, heat may also be generated by magnetic hysteresis losses in the susceptor, i.e. by the varying orientation of magnetic dipoles in the magnetic material as a result of their alignment with the varying magnetic field. In inductive heating, as compared to heating by conduction for example, heat is generated inside the susceptor, allowing for rapid heating. Further, there need not be any physical contact between the inductive heater and the susceptor, allowing for enhanced freedom in construction and application.

(48) The induction heating assembly of the example device **100** comprises a susceptor arrangement **132** (herein referred to as “a susceptor”), a first inductor coil **124** and a second inductor coil **126**. The first and second inductor coils **124**, **126** are made from an electrically conducting material. In this example, the first and second inductor coils **124**, **126** are made from Litz wire/cable which is wound in a helical fashion to provide helical inductor coils **124**, **126**. Litz wire comprises a plurality of individual wires which are individually insulated and are twisted together to form a single wire. Litz wires are designed to reduce the skin effect losses in a conductor. In the example device **100**, the first and second inductor coils **124**, **126** are made from copper Litz wire which has a rectangular cross section. In other examples the Litz wire can have other shape cross sections, such as circular.

(49) The first inductor coil **124** is configured to generate a first varying magnetic field for heating a first section of the susceptor **132** and the second inductor coil **126** is configured to generate a second varying magnetic field for heating a second section of the susceptor **132**. In this example, the first inductor coil **124** is adjacent to the second inductor coil **126** in a direction along the longitudinal axis **134** of the device **100** (that is, the first and second inductor coils **124**, **126** do not overlap). The susceptor arrangement **132** may comprise a single susceptor, or two or more separate susceptors. Ends **130** of the first and second inductor coils **124**, **126** can be connected to the PCB **122**.

(50) It will be appreciated that the first and second inductor coils **124**, **126**, in some examples, may have at least one characteristic different from each other. For example, the first inductor coil **124** may have at least one characteristic different from the second inductor coil **126**. More specifically, in one example, the first inductor coil **124** may have a different value of inductance than the second inductor coil **126**. In FIG. 2, the first and second inductor coils **124**, **126** are of different lengths such that the first inductor coil **124** is wound over a smaller section of the susceptor **132** than the second inductor coil **126**. Thus, the first inductor coil **124** may comprise a different number of turns than the second inductor coil **126** (assuming that the spacing between individual turns is substantially the same). In yet another example, the first inductor coil **124** may be made from a different material to the second inductor coil **126**. In some examples, the first and second inductor coils **124**, **126** may be substantially identical.

(51) In this example, the first inductor coil **124** and the second inductor coil **126** are wound in opposite directions. This can be useful when the inductor coils are active at different times. For example, initially, the first inductor coil **124** may be operating to heat a first section of the article **110**, and at a later time, the second inductor coil **126** may be operating to heat a second section of the article **110**. Winding the coils in opposite directions helps reduce the current induced in the inactive coil when used in conjunction with a particular type of control circuit. In FIG. 2, the first inductor coil **124** is a right-hand helix and the second inductor coil **126** is a left-hand helix. However, in another embodiment, the inductor coils **124**, **126** may be wound in the same direction, or the first inductor coil **124** may be a left-hand helix and the second inductor coil **126** may be a right-hand helix.

(52) The susceptor **132** of this example is hollow and therefore defines a receptacle within which aerosol generating material is received. For example, the article **110** can be inserted into the susceptor **132**. In this example the susceptor **120** is tubular, with a circular cross section.

(53) The device **100** of FIG. 2 further comprises an insulating member **128** which may be generally tubular and at least partially surround the susceptor **132**. The insulating member **128** may be constructed from any insulating material, such as plastic for example. In this particular example, the insulating member is constructed from polyether ether ketone (PEEK). The insulating member **128** may help insulate the various components of the device **100** from the heat generated in the susceptor **132**.

(54) The insulating member **128** can also fully or partially support the first and second inductor coils **124**, **126**. For example, as shown in FIG. 2, the first and second inductor coils **124**, **126** are positioned around the insulating member **128** and are in contact with a radially outward surface of the insulating member **128**. In some examples the insulating member **128** does not abut the first and second inductor coils **124**, **126**. For example, a small gap may be present between the outer surface of the insulating member **128** and the inner surface of the first and second inductor coils **124**, **126**.

(55) In a specific example, the susceptor **132**, the insulating member **128**, and the first and second inductor coils **124**, **126** are coaxial around a central longitudinal axis of the susceptor **132**.

(56) FIG. 3 shows a side view of device **100** in partial cross-section. The outer cover **102** is present in this example. The rectangular cross-sectional shape of the first and second inductor coils **124**, **126** is more clearly visible.

(57) The device **100** further comprises a support **136** which engages one end of the susceptor **132** to hold the susceptor **132** in place. The support **136** is connected to the second end member **116**.

(58) The device may also comprise a second printed circuit board **138** associated within the control element **112**.

(59) The device **100** further comprises a second lid/cap **140** and a spring **142**, arranged towards the distal end of the device **100**. The spring **142** allows the second lid **140** to be opened, to provide access to the susceptor **132**. A user may open the second lid **140** to clean the susceptor **132** and/or the support **136**.

(60) The device **100** further comprises an expansion chamber **144** which extends away from a proximal end of the susceptor **132** towards the opening **104** of the device. Located at least partially within the expansion chamber **144** is a retention clip **146** to abut and hold the article **110** when received within the device **100**. The expansion chamber **144** is connected to the end member **106**.

(61) FIG. **4** is an exploded view of the device **100** of FIG. **1**, with the outer cover **102** omitted.

(62) FIG. **5A** depicts a cross section of a portion of the device **100** of FIG. **1**. FIG. **5B** depicts a close-up of a region of FIG. **5A**. FIGS. **5A** and **5B** show the article **110** received within the susceptor **132**, where the article **110** is dimensioned so that the outer surface of the article **110** abuts the inner surface of the susceptor **132**. This ensures that the heating is most efficient. The article **110** of this example comprises aerosol generating material **110a**. The aerosol generating material **110a** is positioned within the susceptor **132**. The article **110** may also comprise other components such as a filter, wrapping materials and/or a cooling structure.

(63) FIG. **5B** shows that the outer surface of the susceptor **132** is spaced apart from the inner surface of the inductor coils **124**, **126** by a distance **150**, measured in a direction perpendicular to a longitudinal axis **158** of the susceptor **132**. In one particular example, the distance **150** is about 3 mm to 4 mm, about 3 mm to 3.5 mm, or about 3.25 mm.

(64) FIG. **5B** further shows that the outer surface of the insulating member **128** is spaced apart from the inner surface of the inductor coils **124**, **126** by a distance **152**, measured in a direction perpendicular to a longitudinal axis **158** of the susceptor **132**. In one particular example, the distance **152** is about 0.05 mm. In another example, the distance **152** is substantially 0 mm, such that the inductor coils **124**, **126** abut and touch the insulating member **128**.

(65) In one example, the susceptor **132** has a wall thickness **154** of about 0.025 mm to 1 mm, or about 0.05 mm.

(66) In one example, the susceptor **132** has a length of about 40 mm to 60 mm, about 40 mm to 45 mm, or about 44.5 mm.

(67) In one example, the insulating member **128** has a wall thickness **156** of about 0.25 mm to 2 mm, 0.25 mm to 1 mm, or about 0.5 mm.

(68) FIG. **6A** depicts the battery support **120** of FIGS. **2** and **4** in more detail. The battery support **120** comprises a main portion **202**, a first end portion **204** and a second end portion **206**. The main portion **202** defines a longitudinal axis **208**, which is parallel to the axis **134** of the device **100**. The first end portion **204** is arranged at a first end of the main portion **202** and the second end portion **206** is arranged at a second end of the main portion **202**. The first and second end portions **204**, **206** extend away from a first side of the main portion **202** in a direction substantially perpendicular to the longitudinal axis **208**.

(69) In this example the battery **118** is shown disconnected from the battery support **120**. The battery **118** can be connected to the battery support **120** by moving the battery **118** towards the battery support **120** in the direction of arrow **200**. When connected to the battery support **120**, the battery **118** is held between the first and second end portions **204**, **206**. For example, a top end **118a** of the battery **118** is received by the first end portion **204**, and a bottom end **118b** of the battery **118** is received by the second end portion **206**.

(70) FIG. **6A** further depicts the PCB **122** engaged with a second side of the main portion **202**. The PCB **122** may be adhered to the main portion **202** or may be connected via another means, such as

friction fit, snap fit, etc. In this example, the PCB **122** defines a longitudinal axis which is parallel to the longitudinal axis **208** of the main portion **202**.

(71) As described above, the aerosol provision device **100** comprises a heater/heating assembly comprising at least one inductor coil **124**, **126**. FIG. **4** depicts the arrangement of the one or more inductor coils **124**, **126** relative to the battery support **120**.

(72) As shown in FIG. **6A**, the battery **118** comprises a first electrically conductive member **224** in contact with a first battery terminal, and a second electrically conductive member **226** in contact with a second battery terminal. The first and second battery terminals can be positive or negative terminals, for example. The electrically conductive members **224**, **226** may be wires or conductive strips which connect the battery **118** to the PCB **122**. The electrically conductive members **224**, **226** generally extend away from the battery **118** and towards the PCB **122**.

(73) FIG. **6B** shows a close up of a portion of FIG. **6A** from a different viewing angle. FIG. **6B** depicts a haptic component **210**, such as a haptic motor, which provides haptic feedback, such as a vibration. The haptic component **210** may provide feedback in response to an event, such as when a user presses the control element **112** and/or when the susceptor **132** has been heated to the correct temperature.

(74) To ensure that the haptic component **210** is electrically insulated from the battery **118** (such as a battery terminal or the electrically conductive member **226**), the device **100** comprises an electrically insulating resilient member **212** arranged between the battery **118** and the haptic component **210**. The resilient member **212** electrically insulates the haptic component from the battery **118** to avoid shorting the battery **118** by acting as a barrier. Thus, when the battery **118** is connected to the battery support **120**, the bottom end **118b** of the battery (which may include a terminal) and/or the electrically conductive member **226** is in contact with at least a portion of the resilient member **212**.

(75) The resilient member **212** may be arranged in a compressed/deformed state, which causes the resilient member **212** to exert a force on the haptic component **210** in the direction of arrow **214**. This biasing force helps transfer vibrations from the haptic component **210** to other components of the device **100**, by ensuring the haptic component **210** is pressed firmly against another component, such as a second PCB **216**.

(76) In this example, the resilient member **212** is made from silicone rubber, which is a good insulator of heat and electricity.

(77) In the example of FIG. **6B**, the resilient member **212** comprises a first portion **212a**, and a second portion **212b** which extends beyond the first portion **212a** and the haptic component **210**. The first portion **212a** abuts the haptic component **210**, and the second portion abuts the battery terminal and/or the electrically conductive member **226**. The first portion **212a** provides insulation as well as the biasing force. The second portion **212b** primarily provides insulation, so can be made thinner than the first portion **212a**.

(78) FIG. **7** depicts a side view of the resilient member **212**, and FIG. **8** depicts a perspective view of the resilient member **212**. As mentioned, the resilient member **212** comprises a first portion **212a**, and a second portion **212b**. The first portion **212a** is shaped to conform to the haptic component **210**, as shown in FIG. **9**.

(79) The second portion **212b** in this example comprises a base **218** which is at least partially surrounded by one or more side walls **220**. The base **218** and the one or more side walls **220** therefore define a recess/receptacle **222** to receive an end **118b** of the battery **118**. The one or more side walls **220** abut the outer surface of the battery **118** to help hold the resilient member **212** in position relative to the battery **118**. The recess/receptacle **222** is shaped to correspond to the shape of the battery **118**.

(80) As shown in FIG. **7**, the first portion **212a** is thicker than the second portion **212b**. The first portion **212a** has a thickness **228**, and the second portion **212b** has a thickness **232**, defined by the base **218**. The side walls **220** have a thickness/height **230**. The thickness of the second portion **212b**

is therefore defined as the minimum distance between the battery **118** and the first portion, so does not include the thickness/height **230** of any side walls. The thickness **232** of the base **218** must be suitable to provide electrical insulation. The thicknesses **228**, **230**, **232** are each measured in a direction parallel to the longitudinal axis of the device **134**, which is also parallel to the axis **208**. The thickness is therefore measured in a direction away from the battery **118** towards the haptic component **210** (i.e. in the direction indicated by arrow **214** of FIG. 6B).

(81) In the example of FIG. 7, the first portion **212a** has a thickness **228** of between about 2 mm and about 3 mm, such as 2.3 mm, and the second portion **212b** has a thickness **232** of between about 0.5 mm and about 2 mm, such as 0.8 mm. The one or more side walls **220** have a thickness/height **230** of between about 1 mm and about 3 mm, such as 1.2 mm. It has been found that these dimensions provide a good balance between providing adequate electrical insulation and a strong biasing force. In certain examples there are no side walls **220** present. The height/thickness **230** of the one or more sidewalls **220** are suitable to maintain the relative position between the battery **118** and the resilient member **212**.

(82) In some examples, as shown in FIG. 7, the resilient member **212** comprises an engagement feature **234**. The engagement feature **234** engages a corresponding engagement feature of the haptic component **210** to help maintain a relative position between the resilient member **212** and the haptic component **210**. In the specific arrangement shown, the resilient member **212** comprises a protrusion **234**, and the haptic component **210** comprises a recess (not visible). The protrusion **234** is received in the recess. In another example (not depicted), the haptic component **210** may comprise a protrusion, and the resilient member **212** may comprise a recess.

(83) FIG. 10 depicts a perspective view of a portion of the battery **118**, the electrically conductive member **226**, and the resilient member **212**. The engagement feature **234** is positioned on an under side of the first portion **212a** and extends away from the first portion **212a** towards the haptic component **210**.

(84) FIG. 10 shows the electrically conductive member **226** positioned between the resilient member **212** and the end of the battery **118b**. In this example, the end of the battery **118b** comprises the battery terminal, to which the electrically conductive member **226** is connected.

(85) The above embodiments are to be understood as illustrative examples of the invention. Further embodiments of the invention are envisaged. It is to be understood that any feature described in relation to any one embodiment may be used alone, or in combination with other features described, and may also be used in combination with one or more features of any other of the embodiments, or any combination of any other of the embodiments. Furthermore, equivalents and modifications not described above may also be employed without departing from the scope of the invention, which is defined in the accompanying claims.

Claims

1. An aerosol provision device, comprising: a haptic component configured to provide haptic feedback; a battery configured to power the haptic component; an electrically insulating resilient member in contact with the haptic component, wherein the resilient member is at least 0.5 mm thick, and less than 6 mm thick; and an outer cover which surrounds and houses the battery, and which is configured to be held by a user, wherein the resilient member is at least partially compressed, thereby to exert a biasing force on the haptic component so as to help transfer vibrations from the haptic component to the outer cover.
2. An aerosol provision device according to claim 1, wherein the resilient member comprises silicone.
3. An aerosol provision device according to claim 2, wherein the silicone is silicone rubber.
4. An aerosol provision device according to claim 1, further comprising an electrically conductive member in contact with a battery terminal, wherein the resilient member is in contact with the

electrically conductive member.

5. An aerosol provision device according to claim 1, wherein the resilient member is positioned between a battery terminal and the haptic component.

6. An aerosol provision device according to claim 1, wherein the resilient member and the haptic component each comprise corresponding engagement features for maintaining a relative position between the resilient member and the haptic component.

7. An aerosol provision device according to claim 1, wherein the biasing force the haptic component away from the resilient member.

8. An aerosol provision device according to claim 1, wherein the resilient member comprises a first portion and a second portion, the first portion being thicker than the second portion, wherein the haptic component is in contact with the first portion.

9. An aerosol provision device according to claim 8, wherein the second portion comprises a recess configured to receive an end of the battery.

10. An aerosol provision device according to claim 1, wherein the resilient member is less than 3 mm thick.

11. An aerosol provision system, comprising: an article comprising aerosol generating material; and an aerosol provision device comprising: a haptic component configured to provide haptic feedback; a battery configured to power the haptic component; an electrically insulating resilient member in contact with the haptic component, wherein the resilient member is at least 0.5 mm thick, and less than 6 mm thick; and an outer cover which surrounds and houses the battery, and which is configured to be held by a user, wherein the resilient member is at least partially compressed, thereby to exert a biasing force on the haptic component so as to help transfer vibrations from the haptic component to the outer cover.
